

EMCH 260

SOLID MECHANICS

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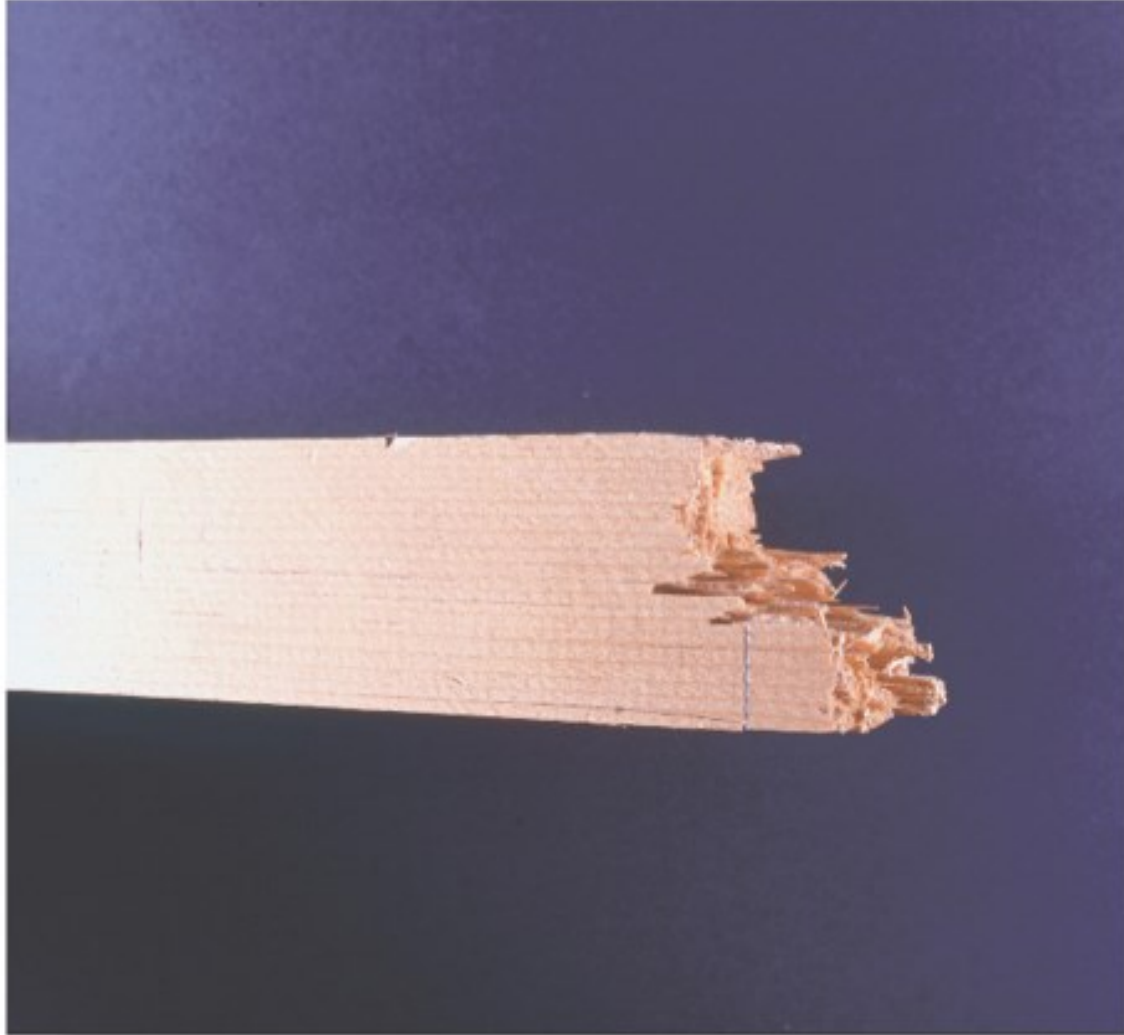
Bending

6

Chapter 6.1

- ✓ Determine stress in members caused by bending
- ✓ Know how to establish shear and moment diagrams for a beam or shaft

APPLICATIONS OF SEVERE BENDING



BEAMS

- Members that are slender and support loadings applied perpendicular to their longitudinal axis are called *beams*



Simply supported beam



Cantilevered beam



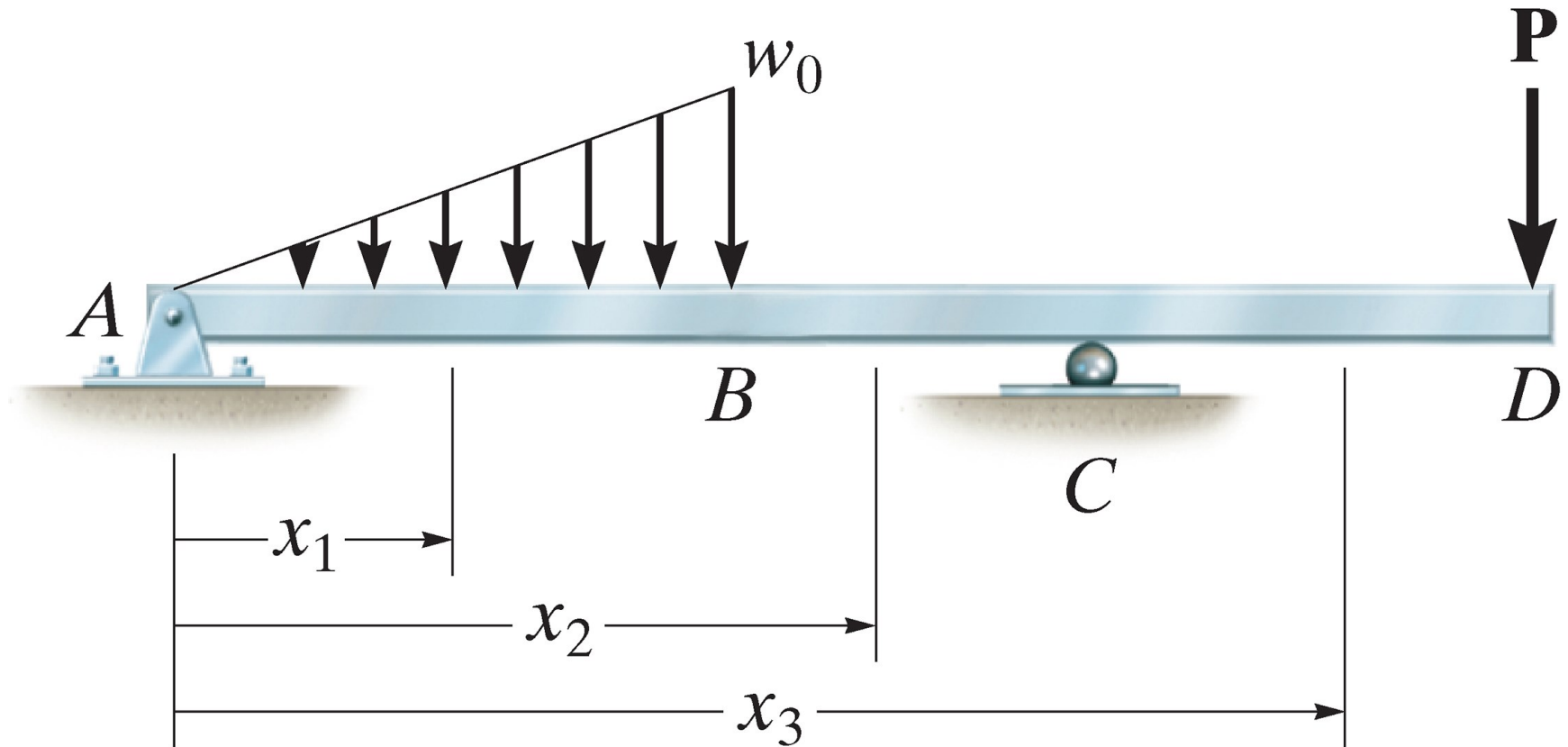
Overhanging beam

INTERNAL SHEAR FORCE AND BENDING MOMENT

- In order to design a beam, it is necessary to determine the maximum shear and moment in the beam
- Express **V** and **M** as functions of arbitrary position x along axis.
- These functions can be represented by graphs called *shear and moment diagrams*
- Engineers need to know the *variation* of shear and moment along the beam to know where to reinforce it

SHEAR AND BENDING MOMENT FUNCTIONS

- Shear and bending-moment functions must be determined for each *region* of the beam between any two discontinuities of loading

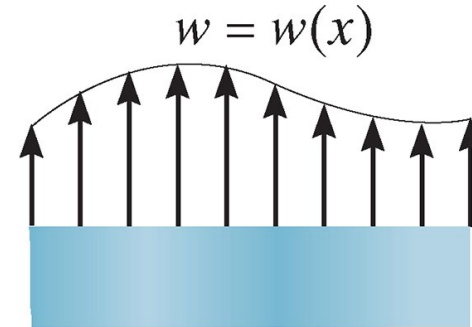


Three different regions should be considered in this case.

BEAM SIGN CONVENTION

Beam sign convention

- Although choice of sign convention is arbitrary, in this course, we adopt the one often used by engineers:



Positive external distributed load



Positive internal shear



Positive internal moment

Beam sign convention

PROCEDURE FOR ANALYSIS

Step 1: Support Reactions

- Determine all reactive forces and couple moments acting on beam
- Resolve all forces into components acting perpendicular and parallel to beam's axis

Step 2: Shear and Moment Functions

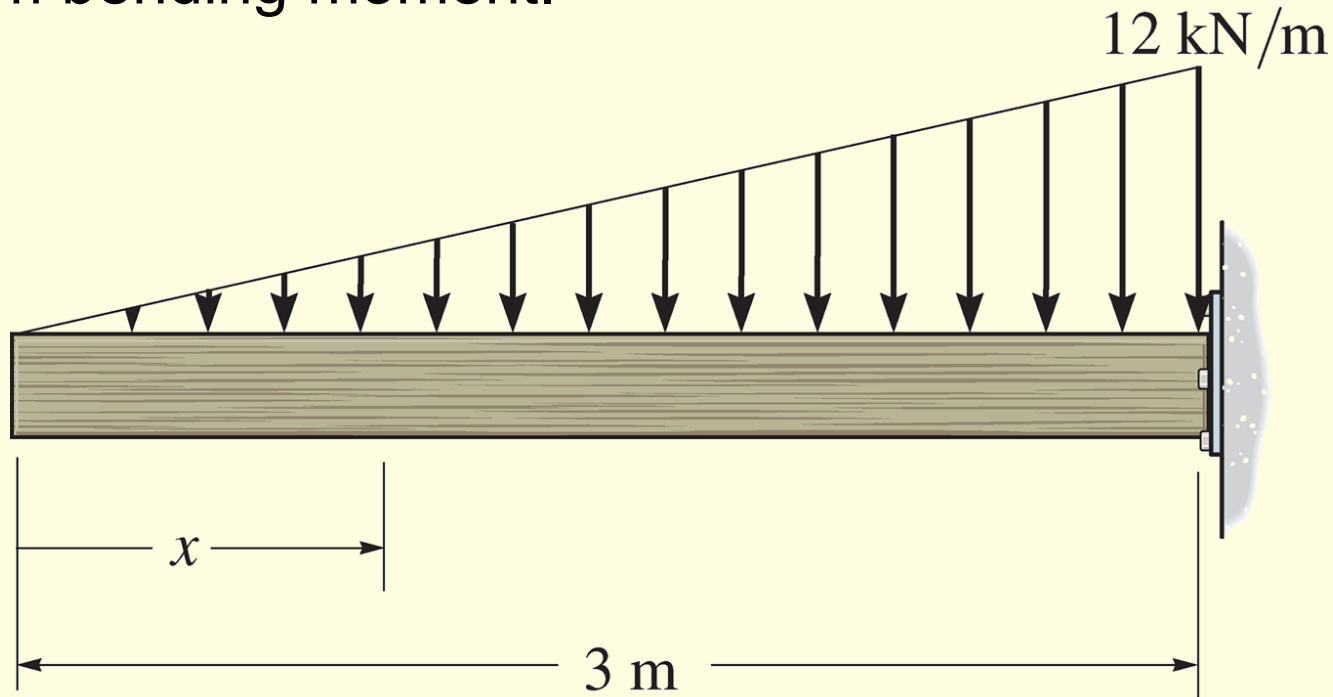
- Specify separate coordinates x having an origin at beam's left end, and extending to regions of beam between concentrated forces and/or couple moments, or where there is no discontinuity of distributed loading

Step 3: Shear and Moment Diagrams

- Section beam perpendicular to its axis at each distance x
- Draw free-body diagram of one segment
- Make sure **V** and **M** are shown acting in positive sense, according to sign convention
- Sum forces perpendicular to beam's axis to get shear force **V**
- Sum moments about the sectioned end of segment to get moment **M**

IN-CLASS EXAMPLE 1

Draw the shear and moment diagrams for the beam shown in the figure below. Determine the position of zero shear and maximum bending moment.



IN-CLASS EXAMPLE 1, cont'd

Solutions

Step 1: Support Reactions

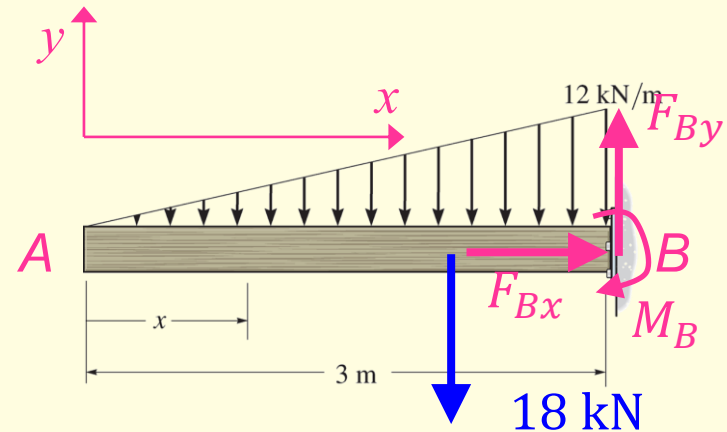
For reactions at joints and distributed loads, refer to Ch.1.2 p. 9-10

The support reactions are shown on the right.

Applying free-body diagram to the entire beam, the force equation of equilibrium gives:

$$\sum F_x = 0 \rightarrow F_{Bx} = 0$$

$$\sum F_y = 0 \rightarrow F_{By} = 18 \text{ kN}$$



The moment equation of equilibrium with respect to point B gives:

$$\begin{aligned} \sum M_B = 0 & \rightarrow 18000 \times 1 = M_B \\ & \rightarrow M_B = 18 \text{ kN} \cdot \text{m} \end{aligned}$$

IN-CLASS EXAMPLE 1, cont'd

Solutions

Step 2: Shear and Moment Functions

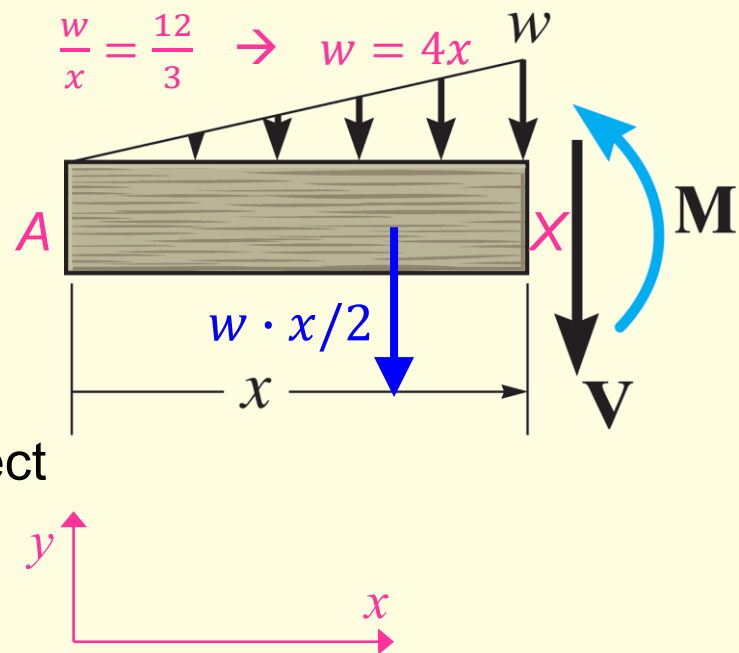
Taking a small segment starting at point A with arbitrary length of x (section position), the free-body diagram of the segment is shown in the right figure.

Applying free-body diagram to the segment, the force equation of equilibrium gives:

$$\begin{aligned}\sum F_y = 0 & \quad \rightarrow \quad w \cdot \frac{x}{2} + V = 0 \\ \rightarrow \quad V &= -w \cdot \frac{x}{2} = -2x^2 \text{ kN} \quad (1)\end{aligned}$$

The moment equation of equilibrium with respect to the **section point X** gives:

$$\begin{aligned}\sum M_X = 0 & \quad \rightarrow \quad M + w \cdot \frac{x}{2} \times \frac{x}{3} = 0 \\ \rightarrow \quad M &= -\frac{2}{3}x^3 \text{ kN} \cdot \text{m} \quad (2)\end{aligned}$$



IN-CLASS EXAMPLE 1, cont'd

Solutions

Step 3: Shear and Moment Diagrams

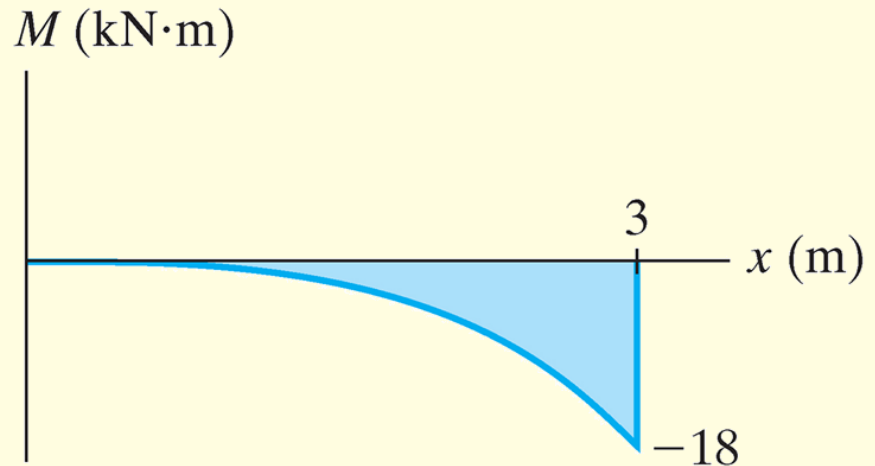
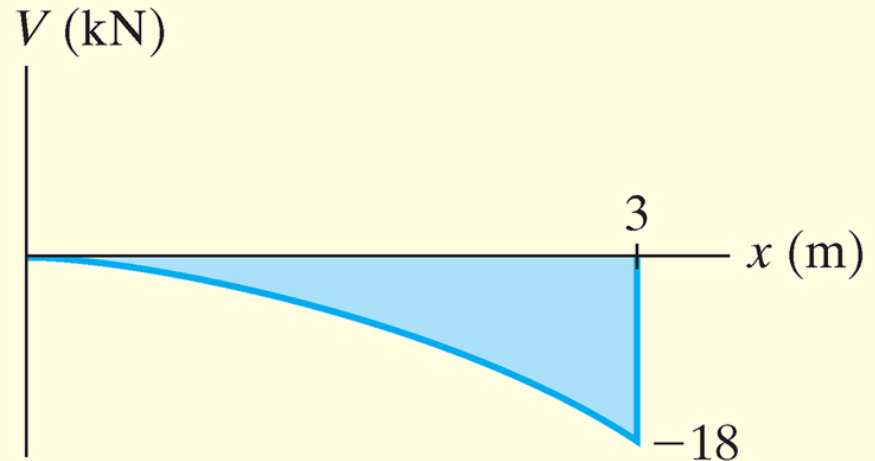
The shear and moment diagrams can be drawn by plotting Eqs. (1) and (2).

The zero shear occurs at:

$$V = 0$$
$$x = 0 \text{ m (left end } A)$$

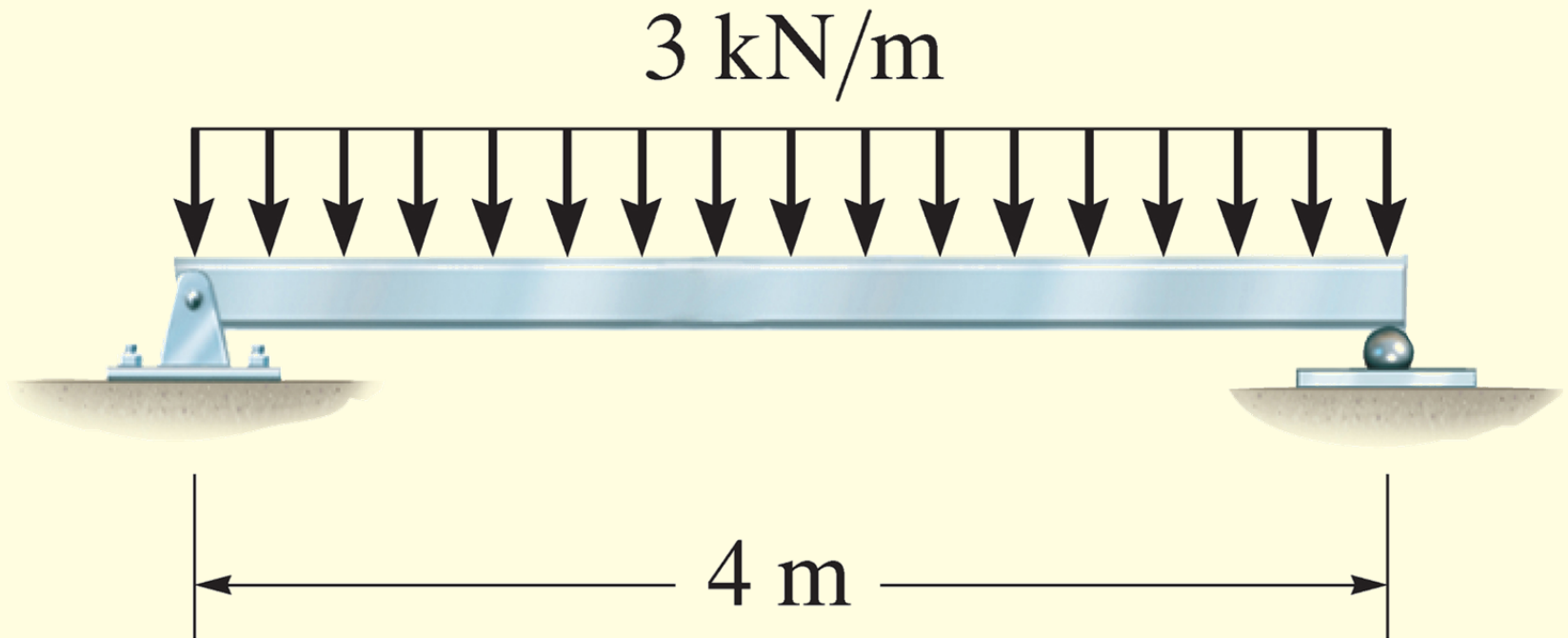
The maximum moment occurs at:

$$M = -18 \text{ kN} \cdot \text{m}$$
$$x = 3 \text{ m (right end } B)$$



IN-CLASS EXAMPLE 2

Draw the shear and moment diagrams for the beam shown in the figure below. Determine the position of zero shear and maximum bending moment.



IN-CLASS EXAMPLE 2, cont'd

Solutions

Step 1: Support Reactions

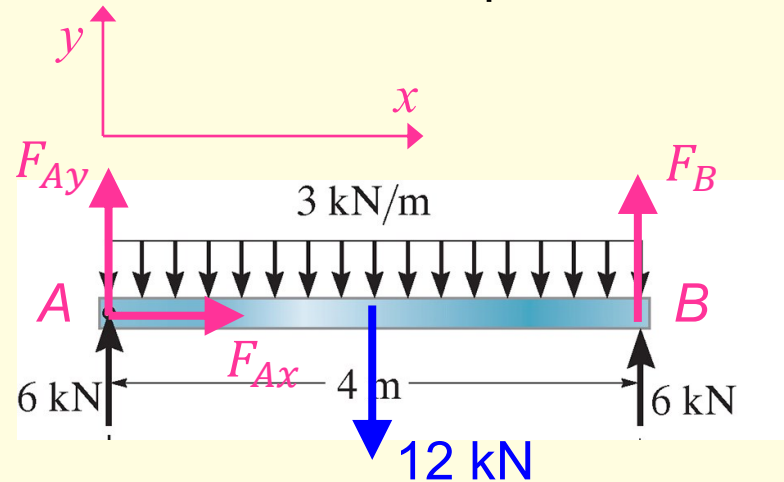
The support reactions are shown on the right.

Applying free-body diagram to the entire beam, the force equation of equilibrium gives:

$$\sum F_x = 0 \rightarrow F_{Ax} = 0$$

$$\sum F_y = 0 \rightarrow F_{Ay} + F_B = 12 \text{ kN}$$

For reactions at joints and distributed loads, refer to Ch.1.2 p. 9-10



The moment equation of equilibrium with respect to point A gives:

$$\begin{aligned} \sum M_A = 0 &\rightarrow 12000 \times 2 - F_B \times 4 = 0 \\ &\rightarrow F_B = 6000 \text{ N} \end{aligned}$$

Then, $F_{Ay} = 6000 \text{ N}$

IN-CLASS EXAMPLE 2, cont'd

Solutions

Step 2: Shear and Moment Functions

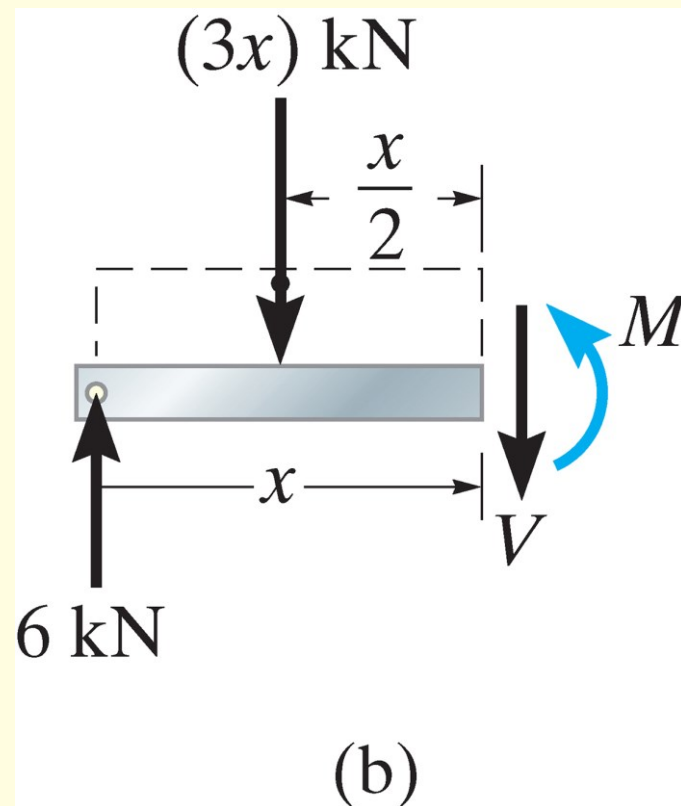
Taking a small segment starting at point A with arbitrary length of x (section position), the free-body diagram of the segment is shown in the right figure.

Applying free-body diagram to the segment, the force equation of equilibrium gives:

$$\begin{aligned}\sum F_y = 0 &\rightarrow 6000 - 3000x - V = 0 \\ &\rightarrow V = 6000 - 3000x \quad (1)\end{aligned}$$

The moment equation of equilibrium with respect to the section gives:

$$\begin{aligned}\sum M = 0 &\rightarrow M - 6000 \cdot x + 3000x \times \frac{x}{2} = 0 \\ &\rightarrow M = 6000x - 1500x^2 \text{ N} \cdot \text{m} \quad (2)\end{aligned}$$



IN-CLASS EXAMPLE 2, cont'd

Solutions

Step 3: Shear and Moment Diagrams

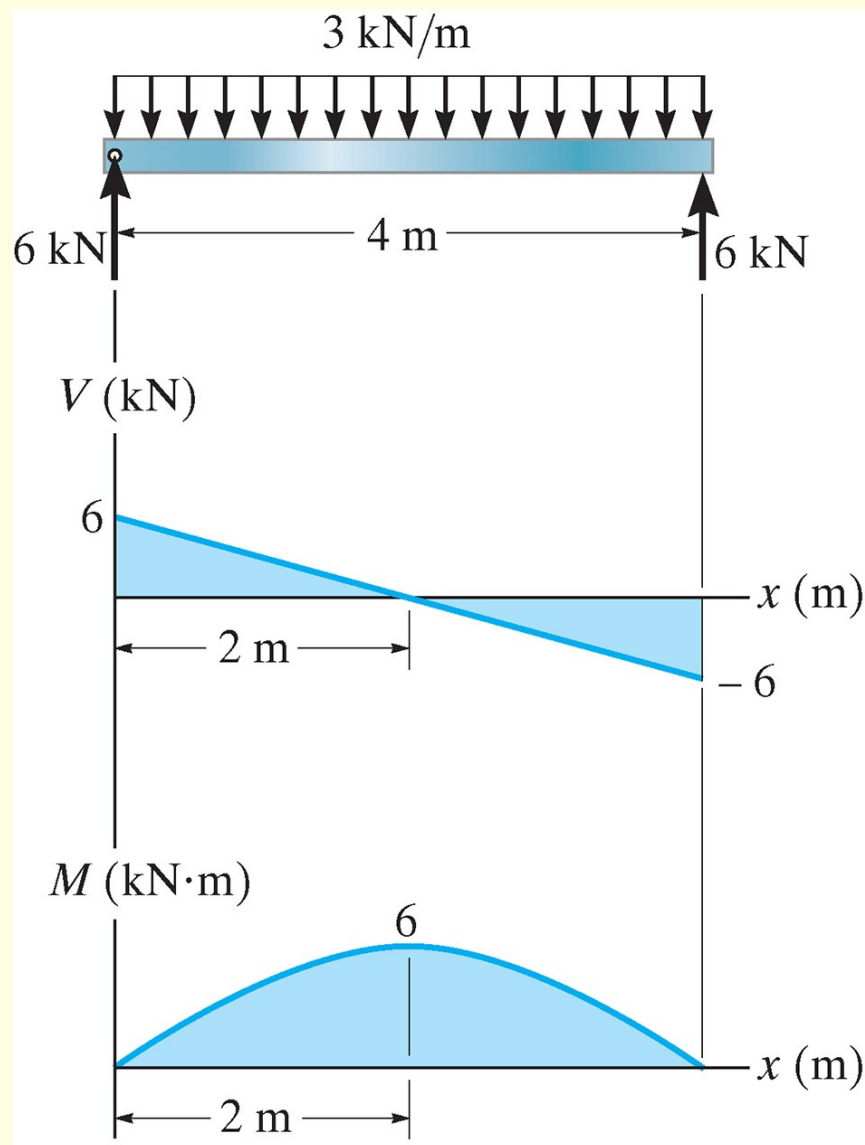
The shear and moment diagrams can be drawn by plotting Eqs. (1) and (2).

The zero shear occurs at:

$$V = 6000 - 3000x = 0$$
$$x = 2 \text{ m (center of the beam)}$$

The maximum moment occurs at:

$$M = 6000x - 1500x^2 = 0$$
$$x = 2 \text{ m (center), } M_{\max} = 6000 \text{ N} \cdot \text{m}$$



RELEVANT PRACTICE

- **Assignment 2 on Blackboard, problem #7 (go to Blackboard under the “Assignments” folder, **Assignment 2 due: 8:30 AM, Friday, Oct. 15th 2021**)**
- **Optional Extra Exercise 2 on Blackboard, problem #5 (go to Blackboard under the “Extra Exercises” folder, **Extra Exercise 2 due: 8:30 AM, Friday, Oct. 15th 2021**)**

OUT-OF-CLASS READING

- **Textbook, pp. 263 – 269**
 - **In particular, Examples 6.1 – 6.4 (pp. 266 – 269)**